

Syed Athar Hussain

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SUMMARY

Software Engineer with 4 years of experience building, deploying, and operating machine learning models and data pipelines in production environments. Skilled in operationalising transformer-based architectures, generative AI, and real-time inference services. Strong background in Python, PyTorch, and distributed systems, with a focus on implementing MLOps best practices to keep ML applications secure, reliable, and scalable.

TECHNICAL SKILLS

- **AI & Machine Learning:** Python, PyTorch, TensorFlow, Scikit-learn, Hugging Face, LLMs, Transformer Architectures, RAG, Tokenization, Embeddings (FAISS)
- **MLOps & Deployment:** Docker, Kubernetes, AWS (SageMaker), MLflow, GitHub Actions (CI/CD), Model Drift Monitoring, Model Evaluation
- **Data Engineering:** Apache Kafka, Apache Spark, Airflow, PostgreSQL, REST APIs, Feature Engineering
- **Software Engineering:** API Integration, Version Control (Git), Pytest, OpenTelemetry, Data Structures, Algorithms

EDUCATION

University College Dublin, Dublin, Ireland Sep 2024 – Aug 2025
Master in Computer Science (Negotiated Learning)

- **Key Modules:** Machine Learning, Data Science, Generative AI, Cloud Computing, Data Visualization, Big Data.

St. Mary's Technical Campus, Kolkata, India Aug 2015 – Jul 2019
Bachelor of Technology in Computer Science

- CGPA: 7.61; Key Coursework: AI, Computer Networks, Data Management.
- Developed an e-commerce application using Laravel, emphasizing scalable backend services and user-focused features.

WORK EXPERIENCE

Fujitsu Consulting India, Pune, India Feb 2022 – Aug 2024
Assistant Application Developer

- Pushed experimental ML models into production. Worked directly with data scientists, software engineers, and product teams to set and hit strict latency budgets and accuracy thresholds.
- Built real-time feature engineering pipelines using Kafka and Airflow to create, validate, and manage large-scale training datasets. This setup guaranteed clean, continuous data delivery for model inference.
- Packaged ML applications in Docker and managed Kubernetes deployments, keeping production uptime around 99.9%.
- Optimised the model-serving architecture by batching requests and tuning Gunicorn worker concurrency, dropping API inference latency from 500ms to under 100ms.
- Set up MLflow to track experiments, monitor long-term model degradation, and manage retraining workflows.
- Led technical discussions to share MLOps best practices across the broader engineering team.
- Wrote Pytest suites for the CI/CD pipelines to catch edge cases before pushing anything to production.

Avant-Grade Technologies, Kolkata, India Mar 2019 – Sep 2021
Full-Stack Developer (Promoted from Junior)

- Developed backend services for an insurance platform using Java, Spring Boot, and PostgreSQL. Wrote custom health checks to meet strict financial regulatory and governance standards.
- Restructured complex database schemas and optimised PostgreSQL queries, reducing overall database load and cutting query execution times by 75%.
- Participated in daily code reviews, technical planning, and testing workflows to ensure the delivery of secure, version-controlled, and maintainable software architecture.

PROJECTS

Error Categorization Engine:

Tech Stack: Python, PyTorch, Hugging Face (BERT), Flask, Docker, Pandas

- Built an offline NLP classification tool that sorts software bugs into 27 root causes. Hosted it entirely on internal infrastructure to guarantee data privacy.
- Fine-tuned a multilingual BERT transformer model in PyTorch using BCEWithLogitsLoss, configuring independent probability thresholds to handle multi-class assignments.
- Wrote custom Pandas scripts for tokenization and text preprocessing, parsing bilingual (English/Japanese) JSON and Excel files for the inference engine.

- Swapped an expensive third-party LLM API for our fine-tuned internal model, cutting inference costs to zero and speeding up manual bug reviews by 1-2x.
- Served the production model via REST APIs using Flask and Gunicorn inside Docker containers.
- Added strict payload size limits and filename validation at the API layer to prevent security exploits on the host machine.

Developer Workflow System:

Tech Stack: Claude, LangGraph, FAISS, Redis, FastAPI, OpenTelemetry

- Built an event-driven GitHub PR reviewer. Used LangGraph supervisor nodes to route incoming code analysis tasks based on how complex they were.
- Set up a RAG pipeline using FAISS vector embeddings to pull relevant code snippets. This kept the LLM within its context window and heavily reduced hallucinations.
- Configured a Redis MCP server to cache repository states across disconnected LLM sessions, which cut API token consumption by 60%.
- Integrated the AI models into backend services via FastAPI, authenticating incoming GitHub webhooks with HMAC SHA-256 signatures to ensure data integrity.
- Added OpenTelemetry to the FastAPI and LangGraph services to monitor model performance, trace execution paths, and debug latency in real-time.

US Stock Data Pipeline Prediction Service:

Tech Stack: PyTorch, Apache Spark, Kafka, AWS SageMaker

- Designed a hybrid neural network combining LSTMs and multi-head self-attention Transformers to process correlations across 120 different equities at the same time.
- Wrote Spark and Kafka data pipelines to calculate technical indicators and normalize 600 feature sequences every minute.
- Bypassed standard AWS SageMaker memory limits during out-of-core distributed GPU training by implementing np.memmap for efficient memory mapping.
- Used Early Stopping and ReduceLROnPlateau during training to monitor model drift and keep predictions stable.
- Achieved over 75% directional accuracy in batch predictions across the evaluation dataset.
- Pushed low-latency model predictions straight to a live trading desk using automated ML workflows.

AWARDS

- **UCD Advantage Award:** Recognised by University College Dublin for participation in co-curricular and community activities.
- **STAR Silver Awards:** Earned twice at Fujitsu for outstanding performance in system reliability and optimisations.
- Secured second position in Science Exhibition organised by National Service Scheme (NSS).
- Awarded First Position in the NSS Wall Magazine competition for the conceptualization and designed a visual narrative focused on global warming and environmental advocacy.

CERTIFICATION & EXTRA-CURRICULAR ACTIVITIES

- **Certifications:**
 - Full Stack Java Development
 - Data Science Foundation
 - Google SRE Fundamentals
- **Participations :**
 - GeeksforGeeks Hackathon: Competed in coding challenges, applying ML and backend skills.
 - HackeRank Netsoc 2024 Contest: Participated in problem-solving events focused on algorithms and data structures.